

for this information present in the pen itself, not the tablet. On older tablets, changing the pressure on the pen nub or pressing a switch changed the properties of the LC circuit, affecting the signal generated by the pen, which modern ones often encode as a digital data stream onto the signal. By using electromagnetic signals, the tablet detects the stylus position without the need for the stylus to even touch the surface. Powering the pen with this signal means that devices used with the tablet never need batteries. Wacom's patents prevent their competitors to employ such techniques.

Active Tablets

The difference in active tablets is that the stylus used contains self-powered electronics that generate and transmit a signal to the tablet. These pens rely on an internal battery rather than the tablet for their power, resulting in a bulkier pen. Eliminating the need to power the pen means that such tablets may listen for pen signals constantly, as they do not have to alternate between transmit and receive modes, which can result in less jitter.

For both technologies, the tablet uses the received signal to determine the distance of the stylus from the surface of the tablet, the tilt (angle from vertical) of the stylus, and as well as other information in addition to the horizontal and vertical positions.

Compared to a resistive or capacitive touch screen, a graphics tablet offers improved precision, with the ability to track an object not touching the tablet, and can gather significant information about the stylus, but at the same time is more expensive, and can only be used with the special stylus or other accessories.

Some tablets, especially inexpensive ones aimed at young children, come with a corded stylus, using technology similar to older RAND tablets, although this design is no longer used on any normal tablets.

Graphics tablets come with accessories too. The initial accessories were special mouse-like pucks designed for CAD applica-